METRIC

MIL-PRF-81298E
28 September 2011

SUPERSEDING
MIL-D-81298D
29 January 1988

PERFORMANCE SPECIFICATION

DYE, LIQUID FOR THE DETECTION OF LEAKS IN AIRCRAFT FUEL SYSTEMS

Reactivated after 28-September-2011 and may be used for new and existing designs and acquisitions.

Comments, suggestions, or questions on this document should be addressed to AFPA/PTPT 2430 C Street, Bldg 70, Area B, Wright-Patterson AFB OH 45433-7632 or e-mailed to AFPET.AFTT@wpafb.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil or https://assist.daps.dla.mil/quicksearch.

AMSC N/A FSC 6820

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:09Z
Check the source to verify that this is the current version before use.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for liquid dyes which are added to aircraft fuel systems to aid in the detection of fuel leaks (see 6.1).

1.2 Classification. Liquid dye will be of the following types as specified (see 6.2.)

Type	I	Red
Type	II	Yellow
Type	III	Yellow/Green Fluorescent

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-83133	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. DEPARTMENT OF LABOR (DOL)

OSHA 29 CFR 1910.1200	Occupational Safety and Health Standards Toxic and Hazardous Substances Hazard Communication

(Copies of these documents are available online at http://www.osha.gov or may be obtained from OSHA Publication Office, Room S-4203, 200 Constitution Avenue NW, Washington, DC 20210.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:09Z
Check the source to verify that this is the current version before use.

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4	Sampling Procedures and Tables for Inspections by Attributes

(Copies of these documents are available online at http://webstore.ansi.org or may be obtained from the American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036. Copies are also available online at http://asq.org or may be obtained from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

ASTM INTERNATIONAL

ASTM D97	Standard Test Method for Pour Point of Petroleum Products (DoD Adopted)
ASTM D1310	Standard Test Methods for Flash Point and Fire Point of Liquids by Tag Open-Cup Apparatus (DoD Adopted)
ASTM D2276	Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling (DoD Adopted)
ASTM D3241	Standard Test Method for Thermal Oxidation Stability of Aviation Turbine Fuels (DoD Adopted)
ASTM D3278	Standard Test Methods for Flash Point of Liquids by Small Scale Closed-Cup Apparatus (DoD Adopted)
ASTM D4057	Standard Practice for Manual Sampling of Petroleum and Petroleum Products (DoD Adopted)
ASTM D4177	Standard Practice for Automatic Sampling of Petroleum and Petroleum Products (DoD Adopted)
ASTM D6304	Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fischer Titration (DoD Adopted)

(Copies of these documents are available online at http://www.astm.org or at ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection in accordance with 4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:09Z
Check the source to verify that this is the current version before use.

3.2 Materials. The dyes shall consist essentially of liquid oil soluble dyes, completely organic and containing no reactive amino groups. The contractor shall certify that no carcinogenic or potentially carcinogenic constituents are present as defined under the Hazard Communications Standard (HCS) (29 CFR 1910.1200). Certificate to this effect shall be made available to the contracting officer or the contracting officer's designated representative. The composition of the dye shall be made available upon request of the using activity. The product shall be entirely suitable for the intended use (see 6.1).

3.3 Chemical and physical requirements. The chemical and physical requirements of the finished dyes shall conform to the requirements listed in section 3, Table I and Table II when tested in accordance with the applicable test methods. Requirements contained herein are not subject to correction for test tolerances.

3.4 Storage stability. The finished dyes shall remain stable and conform to the requirements in section 3, Table I and Table II as applicable for at least one year when tested in accordance with 4.4.5.

3.5 Thermal stability. In addition to the requirements of Table I and Table II, Type I (Red), Type II (Yellow) and Type III (Yellow/Green Fluorescent) dyes shall not degrade the thermal stability of MIL-DTL-5624 or MIL-DTL-83133 fuels, when tested in accordance with ASTM D3241 (see 4.4.2). This test is applicable only for the first article sample.

3.6 Color. The liquid dye shall be Type I (Red), Type II (Yellow) or Type III (Yellow/Green Fluorescent) as specified (see 6.2).

3.7 Workmanship. When examined visually, the liquid dye shall be completely homogeneous material free from any insoluble matter.

TABLE I. Physical requirements and test methods for Type I Red and Type II Yellow Dye.

Requirements	Min	Max	Test Method
Moisture, mass percent		0.08	D6304
Flash point, open cup, °C	32		D1310
Flash point, closed cup, °C	25		D3278
Pour point, °C		-18	D97
Color value (absorptivity)			1/
Type I (Red)	30		
Type II (Yellow)	24		
Wavelength at maximum absorption, nanometers (nm)			1/
Type I (Red)	510	522	
Type II (Yellow)	423	442	
Contamination of aircraft fuel, mg/L			D2276 2/
Type I (Red)		2.0	
Type II (Yellow)		1.0	
Notes: 1/ See 4.4.3 for test procedure 2/ See 4.4.1 for test procedure			

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:09Z
Check the source to verify that this is the current version before use.

TABLE II. Physical requirements and test methods for Type III Yellow/Green Fluorescent Dye.

Requirements	Min	Max	Test Method
Moisture, mass percent		0.08	D6304
Flash point, open cup, °C	43		D1310
Flash point, closed cup, °C	40		D3278
Pour point, °C		-18	D97
Percent transmittance at 525 nanometers (nm)			
a. Dilution ratio dye: Xylene = 1: 1250		80	1/
b. Dilution ratio dye: Xylene = 1: 833		71	1/
Contamination of aircraft fuel, mg/L		1.0	D2276 2/
Notes: 1/ See 4.4.4 for test procedure 2/ See 4.4.1 for test procedure			

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. Test requirements of section 3 shall be complied with prior to bid on any product required under this specification.

4.2.1 First article sample**.** When a contractor has not previously furnished material under this specification, one liter of the liquid dye from the first batch that represents the product offered to the Government under any contract shall be forwarded to the Aerospace Fuels Laboratory, AFPA/PTPLA, 2430 C Street, Building 70 Area B, Wright Patterson AFB OH 45433-7632. Test data which show the results of tests required by section 3 shall accompany the sample submitted for first article testing.

4.3 Conformance inspection. Testing for individual lots shall consist of all requirements specified in section 3, except storage stability. Quality conformance inspection shall include the test requirements herein.

4.3.1 Inspection lots. For acceptance purposes, individual lots shall be examined as specified herein and subjected to all requirements cited in section 3.

4.3.1.1 Bulk lot. A bulk lot shall consist of an indefinite quantity of a homogeneous mixture of liquid dye offered for acceptance in a single, isolated container; or manufactured in a single plant run (not to exceed 24 hours), through the same processing equipment, with no change in the ingredient materials.

4.3.1.2 Packaged lot. A packaged lot shall consist of an indefinite number of one gallon containers or smaller unit packages of identical size and type, offered for acceptance, and that have been filled with a homogeneous mixture of liquid dye of one color, manufactured

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:09Z
Check the source to verify that this is the current version before use.

in a single plant run (not to exceed 24 hours), through the same processing equipment, with no change in the ingredient materials.

4.3.2 Sampling Plans.

4.3.2.1 Sampling for the examination of filled containers. Take a random sample of filled containers from each lot in accordance with ANSI/ASQC Z1.4.

4.3.2.2 Sampling for tests. Take samples from bulk or packaged lots for tests in accordance with ASTM D4057 or ASTM D4177, or both, except where individual test procedures contain specific sampling instructions.

4.4 Inspection test methods. Tests to determine conformance to chemical and physical requirements shall be conducted in accordance with ASTM Standards, using the applicable methods as listed in Table I, Table II and 4.4.1 through 4.4.5.

4.4.1 Particulate contamination. Aviation turbine fuel conforming to MIL-DTL-5624, grade JP-5, shall be filtered through a 0.45 micrometer membrane manufactured by Millipore Filter Corporation, 225 Ashby Road, Bedford, MA 01730 or equivalent. Dye the prefiltered fuel in the ratio 57 g of dye to 378 L of fuel, assuring a uniform mixture. The contamination resulting from the dissolved liquid dye shall be determined in accordance with ASTM D2276. Repeat this test using prefiltered fuel conforming to MIL-DTL-83133, grade JP-8.

4.4.2 Thermal stability. Fuel conforming to MIL-DTL-5624, grade JP-5, or MIL-DTL-83133, grade JP-8, shall be blended in ratios 45 g of dye in 378 L of fuel. When testing in accordance with ASTM D3241, the thermal stability of the dyed JP-5 or JP-8 fuels shall meet MIL-DTL-5624 or MIL-DTL-83133 requirements, as applicable.

4.4.3 Color value and wavelength of maximum absorption. Accurately weigh to 0.0001 g a 0.3800 g to 0.4000 g sample of red dye or a 0.4200 g to 0.4400 g sample of yellow dye and transfer to a 100 mL volumetric flask. Any amount within the given ranges may be taken. Dilute to the mark with spectral grade xylene (equivalent to American Chemical Society Reagent Grade, suitable for use in spectrophotometry). Agitate until the solution is uniform. Using a class A pipette, transfer 2.00 mL of this solution into a 250 mL volumetric flask and dilute to a mark with iso-octane. Record the visible spectrum of the final solution in a 1.000 cm cell versus that of iso-octane employing a spectrophotometer which has been checked for wavelength accuracy and photometric accuracy by standard procedure. Determine the wavelength of maximum absorbance by inspection of the curve. Calculate the absorptivity of a liquid dye using expression:

$$a = \frac{A}{(bc)}$$

Where:

a = absorptivity

A = absorbance of the solution at the wavelength of maximum absorbance

b = cell thickness in centimeters (cm)

c = concentration of the liquid dye in iso-octane in grams per liter. If the recommended procedure is followed then the concentration of the measure dye solution in grams per liter is equal to the sample weight in grams divided by 12.5 liters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:09Z
Check the source to verify that this is the current version before use.

4.4.4 Percent transmittance measurement. Accurately weigh to 0.0001 g approximately 1.000 gram fluorescent dye in a 250 mL glass beaker. Dilute the above dye 1250 times (by weight) with spectral grade xylene (equivalent to American Chemical Society Regent Grade, suitable for use in spectrophotometry). Agitate until the solution is uniform. Fill a 2.54 cm diameter glass tube cell with this solution. Measure the percent Transmittance of this solution using a spectrophotometer at 525 nm (nanometer). Repeat the above procedure, but this time, dilute the fluorescent dye 833 times with xylene.

4.4.5 Storage stability. Store a one liter sample of the liquid dye in its original container at a laboratory room temperature of 25 °C ± 3 °C for 12 months. After 12 months, shake the can vigorously by hand and examine visually for evidence of product deterioration or can corrosion. After the visual examination, subject the stored sample to all examinations and tests of this specification.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended Use. The liquid dyes covered by this specification are intended to aid in the detection of aircraft fuel system leaks by coloring aircraft fuels. Fuel system leaks accrue in military aircraft due to the extreme stress conditions the airframes encountered. Commercial aircraft are not subject to these stress conditions. There is no currently available non-government standard which fulfills the requirements of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and color of dye required. (see 3.6).
c. Packaging requirements (see 5.1).
d. Quantity required and size containers desired.
e. One copy of the certificate of analysis that lists those items in Table I and Table II as applicable should be forwarded to the acquisition activity.

6.3 Conversion of metric units. Units of measure have been converted to the International System of Units (Metric) in accordance with IEEE/ASTM SI10.

6.4 Safety. Adequate ventilation should be provided to remove the highly flammable and toxic solvent vapors.

6.5 Material Safety Data Sheets. Contracting officers will identify those activities that require copies of completed Material Safety Data Sheets (MSDS) prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:09Z
Check the source to verify that this is the current version before use.

6.6 Subject term (key word) listing.

Aviation Turbine fuel
JP-8
JP-8 + 100
JP-5

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – AV
Navy – AS
Air Force – 68
DLA – GS

Review activities:
Army – EA, AT, MR, SM
Navy – SH, CG
Air Force – 11, 19, 99
DLA – PS

Preparing activity:
Air Force – 68
(Project 6820-2011-001)

Note: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information using the ASSIST Online database at https://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:09Z
Check the source to verify that this is the current version before use.